

DormGuard

User Manual

Monitor a dormitory door, receive alerts when it stays open, control the DormGuard hardware remotely, and review door activity logs.

Covers the Android application, Google Apps Script / Google Sheets integration, and the ESP32-based DormGuard hardware.

Document version reflects the repository's current main branch
(Android UI, application logic, notification service, and ESP32 firmware)

Table of Contents

- 1. What DormGuard Does**
- 2. Before Using DormGuard**
- 3. First-Time App Setup**
- 4. Home Screen**
- 5. Door Alert System**
- 6. Phone Notifications**
- 7. Device Controls**
- 8. Automatic Lamp**
- 9. Logs**
- 10. View Logs in Google Sheets**
- 11. Export Logs**
- 12. Monitoring Frequency**
- 13. Appearance Settings**
- 14. Accessibility Settings**
- 15. Backup and Restore**
- 16. Recommended Everyday Setup**
- 17. Troubleshooting**
- 18. Important Alert Behavior**
- 19. Quick Reference**

1. What DormGuard Does

DormGuard consists of an Android app connected through Google Apps Script and Google Sheets to an ESP32-based hardware device. The ESP32 monitors the door and motion sensor, while the app displays status and provides remote controls.

DormGuard can:

- Display the current door status
- Show the last time the door was opened
- Detect when the door has remained open for too long
- Trigger a hardware alert using the buzzer and NeoPixel LEDs
- Send an alert notification to the Android phone
- Stop an active alert remotely
- Automatically stop an alert after 20 seconds
- Automatically control a lamp when motion is detected
- Enable or disable the hardware LED, buzzer, and lamp
- Record door open/close events
- View logs stored in Google Sheets
- Export all records or the latest 50 records as CSV
- Adjust monitoring frequency
- Change appearance and accessibility settings

2. Before Using DormGuard

The DormGuard hardware must already be installed and configured before using the app.

ESP32 Requirements

- Wi-Fi connection
- Door / reed sensor
- Ultrasonic motion sensor
- Buzzer
- NeoPixel LED strip

The ESP32 connects to Google Apps Script endpoints for logging and remote control.

Phone Requirements

- Internet access, to communicate with the Google Apps Script endpoint
- Android notification permission — required only if you want phone alerts

3. First-Time App Setup

On the Home screen, tap **Setup DormGuard**. You will be given two setup methods: **Manual Setup** and **QR Setup**. The QR option can scan a setup QR code using the phone camera or from an image in the gallery.

Option A — QR Setup

- 1 Tap **Setup DormGuard**.
- 2 Select **QR Setup**.
- 3 Allow camera access if Android asks.
- 4 Point the camera at the DormGuard setup QR code.
- 5 Wait for the QR code to be recognized.
- 6 Confirm/save the imported configuration.

You can also switch to **Gallery** and select an image containing the QR code.

Option B — Manual Setup

Go to:

```
Settings → DormGuard Configurations → Google Sheet API
```

The configuration contains three important pieces:

Field	Description
Door Logs	Google Apps Script Deployment ID used for door logging. This is the most important field — the app uses it to retrieve door history and current status.
Sheets Link	The Google Sheets URL. Optional but recommended; lets the app open the underlying spreadsheet directly.
Controls	Google Apps Script endpoint/Deployment ID used for device controls. Required if you want to remotely control the hardware from the app.

After entering the values, tap **Save**.

4. Home Screen

The Home screen is the main monitoring screen. It contains several important sections.

Door Status

The Door Status card shows the latest state reported by the DormGuard device:

- **OPEN** — the door is open
- **CLOSED** — the door is closed

The app periodically retrieves the latest data from the configured API.

Last Opened At

Shows the most recently recorded door-opening time and date. Example:

14:35 20 Aug 2026

Time

Displays the current phone time.

Lamp / Occupancy Indicator

The lamp indicator provides information about the hardware's lamp and motion state. The device distinguishes between four states, determined using the ultrasonic sensor and the lamp state:

State	Meaning
Light off + not occupied	Lamp off, no motion
Light off + occupied	Motion detected, lamp off
Light on + not occupied	Lamp on, no motion
Light on + occupied	Lamp on, motion detected

5. Door Alert System

DormGuard automatically starts an alert when the door remains open for more than approximately 5 seconds. The hardware checks the door continuously and starts the alert after this configured threshold.

When the alert becomes active:

- The hardware buzzer can beep
- The NeoPixel LEDs can blink red
- The app can show an active alert
- The phone can vibrate, if alert notifications are enabled

Stopping an Alert

When an alert is active, the Home screen's Alert Control changes to **Disable Alert**. Tap it to stop the alert. The app sends a remote stop command to the DormGuard device. The device then suppresses the current alert until the door is closed and subsequently reopened.

Automatic Alert Timeout

If nobody disables the alert manually, the hardware automatically stops the alert after approximately 20 seconds. The alert is then suppressed until the door is closed and opened again.

6. Phone Notifications

To receive Android notifications:

- 1 Open **Settings**.
- 2 Find **Notification**.
- 3 Enable it.
- 4 Grant Android notification permission when requested.

DormGuard uses an Android foreground service for door monitoring and creates a high-importance **Door Alerts** notification channel.

Important: If notification permission is denied, phone notifications are disabled.

7. Device Controls

Go to:

Settings → DormGuard Device

Enable **Device Controls**. This allows the app to communicate with the ESP32 through the configured control endpoint.

You can control:

Control	Description
LED Blink	Controls whether the NeoPixel LED is used for the door alert. When enabled, an active alert produces a blinking red LED effect.
Buzzer	Controls the audible alert. When enabled, the buzzer sounds during an active door alert.
Lamp	Controls the automatic lamp functionality. When enabled, motion detected by the ultrasonic sensor can turn the lamp on. The ESP32 automatically turns the lamp off after approximately 10 seconds without detected motion.

8. Automatic Lamp

The lamp operates using the motion sensor. When motion is detected:

- 1 The ESP32 detects a nearby object within its configured detection distance.
- 2 The lamp turns on if the Lamp setting is enabled.
- 3 When motion stops, the device waits approximately 10 seconds.
- 4 If no new motion is detected, the lamp turns off.

The current firmware uses approximately **32 cm** as the ultrasonic detection threshold.

9. Logs

Open the **Logs** tab.

The log screen displays recorded door events with:

- Time
- Date
- Door state

The ESP32 sends door state changes to Google Apps Script, which stores the records in Google Sheets. Typical records include **OPEN** and **CLOSED**. The system can also record the lamp/motion state associated with the event.

10. View Logs in Google Sheets

On the Logs screen, tap the **Sheets** button near the page title. DormGuard will open the configured Google Sheets link. This is useful when you need to inspect the underlying database rather than the simplified app log.

11. Export Logs

Under **Download Logs**, there are two options.

Option	Result
Full Records	Exports the complete set of records returned by the configured API as full_records.csv
Latest	Exports the newest 50 records as latest_records.csv

The CSV contains: Time, Date, Door.

Android will ask you to confirm saving the file.

12. Monitoring Frequency

Go to:

Settings → Monitoring Frequency

There are five modes:

Mode	Update Interval
Low	10 seconds
Medium	8 seconds
High	5 seconds
Max	2 seconds
Smart	Automatically adjusts

- **Low** uses less battery/data
- **Max** provides the fastest app-side status updates but uses more network activity
- **Smart** normally uses slower polling and increases frequency when activity is detected

For normal use, **Medium** or **Smart** is appropriate.

13. Appearance Settings

DormGuard provides:

- Color Palette
- Dynamic Color
- Dark Mode

These affect the appearance of the application and do not change the physical DormGuard device.

14. Accessibility Settings

Under **Accessibility**, you can enable:

Setting	Effect
Disable Animation	Reduces/removes interface animations
High Contrast	Increases visual contrast for improved readability

Setting	Effect
Disable Haptic	Disables vibration/haptic feedback from the application

15. Backup and Restore

Under:

Settings → Data Management

you can find:

- Encrypt Backup
- Backup
- Restore
- Reset All Data
- Enable **Encrypt Backup** if you want the backup protected before creating it
- Use **Backup** before changing or resetting your configuration
- Use **Restore** to import a previous configuration

Caution: Reset All Data should only be used when you intentionally want to clear the application's stored configuration/data.

16. Recommended Everyday Setup

For normal dormitory monitoring:

- 1 Turn on Device Controls.
- 2 Turn on LED Blink.
- 3 Turn on Buzzer.
- 4 Turn on Lamp, if the automatic lamp is required.
- 5 Turn on Notification.
- 6 Turn on Alert.
- 7 Set Monitoring Frequency to Smart or Medium.
- 8 Leave DormGuard running so Android can maintain its monitoring service.

When leaving the room, close the door and verify that the app displays **CLOSED**. When the door is opened and remains open for more than approximately 5 seconds, the alert system should activate.

17. Troubleshooting

Door status does not update

Check:

- 1 The ESP32 has Wi-Fi.
- 2 The Door Logs API is configured correctly.
- 3 The Google Apps Script deployment is accessible.
- 4 Internet access is available on the phone.
- 5 Monitoring Frequency is not set to an extremely slow mode.

Remote controls do nothing

Check:

- 1 Device Controls is enabled.
- 2 The Controls API is configured.
- 3 The ESP32 has Internet access.
- 4 The control Google Apps Script deployment is working.

The ESP32 polls its control endpoint approximately every 2 seconds when connected.

No phone notification

Check:

- 1 DormGuard's Notification setting.
- 2 Android notification permission.
- 3 Android battery/background restrictions.
- 4 The app's monitoring service is running.

Buzzer does not sound

Check:

- 1 Buzzer is enabled.
- 2 Device Controls is enabled.
- 3 The ESP32 is connected to the control API.
- 4 The door has actually remained open long enough to trigger the alert.

LED does not blink

Check:

- 1 LED Blink is enabled.
- 2 The ESP32 is connected.
- 3 The door is open long enough to trigger the alert.

Lamp does not turn on

Check:

- 1 Lamp is enabled.
- 2 Motion is within the sensor's detection range.
- 3 The ultrasonic sensor is connected correctly.
- 4 The ESP32 is running normally.

18. Important Alert Behavior

The DormGuard alert is based on the physical ESP32 state, not solely on the Android application. This behavior is implemented directly in the ESP32 firmware.

1	Door opens
2	Wait ~5 seconds
3	Alert activates
4	Buzzer/LED operate according to settings
5	User disables alert OR ~20 seconds passes
6	Alert stops
7	Alert remains suppressed
8	Door closes
9	System resets
10	Next door opening can trigger a new alert

19. Quick Reference

Task	Where
Check door	Home → Door Status
Check last opening	Home → Last Opened At
Stop active alert	Home → Alert Control
View history	Logs
Open Google Sheets	Logs → Sheets
Export all logs	Logs → Full Records
Export latest 50	Logs → Latest
Enable hardware controls	Settings → Device Controls
Enable/disable LED	Settings → LED Blink
Enable/disable buzzer	Settings → Buzzer
Enable/disable lamp	Settings → Lamp
Phone notifications	Settings → Notification
Door alert monitoring	Settings → Alert
Change polling speed	Settings → Monitoring Frequency
Dark mode	Settings → Dark Mode
Accessibility	Settings → Accessibility
Configure APIs	Settings → Google Sheet API
Backup	Settings → Backup
Restore	Settings → Restore
Clear app data	Settings → Reset All Data

This manual is based on the repository's current main branch, including the Android UI, JavaScript application logic, Android notification service setup, and ESP32 firmware behavior.